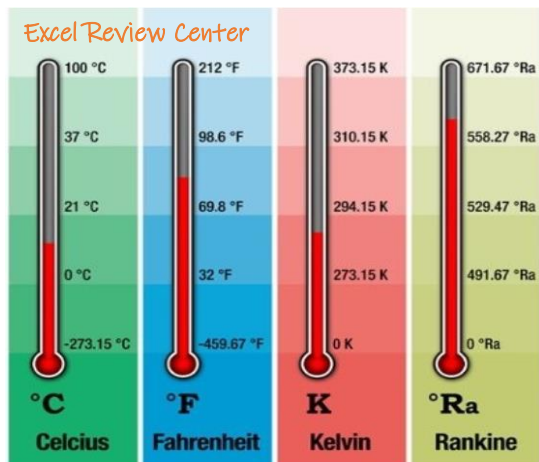




Temperature Scales

continuation...



	Celsius	Fahrenheit	Kelvin	Rankine
Boiling point of water	100°	212°	373.15	671.67°
Freezing point of water	0°	32°	273	492°
Absolute zero	- 273.15°	-459.67°	0	0°

	To Fahrenheit	To Kelvin
Celsius	$F = \frac{9}{5}C + 32$	$K = C + 273.15$

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	To Celsius	To Rankine
Fahrenheit	$C = \frac{5}{9}(F - 32)$	$R = F + 459.67$

Thermal Expansion

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There are two types of expansion, either linear or volume. For solids, both expansion types are significant. For liquids and for gases, it's usually the volume expansion that is meaningful.

Linear Expansion

Linear expansion is the change in length per degree of temperature change.

$$\Delta L = \alpha L \Delta T$$

where:

 α = coeff. of linear expansion ($^{\circ}\text{C}^{-1}$)

L = initial length (m)

 ΔL = change in length (m) ΔT change in temperature ($^{\circ}\text{C}$)

Volume Expansion

Volume expansion is the change in volume per degree of temperature change.

$$\Delta V = \beta V \Delta T$$

where:

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 β = coeff. of linear expansion ($^{\circ}\text{C}^{-1}$)V = initial volume (m^3) ΔV = change in volume (m^3) ΔT change in temperature ($^{\circ}\text{C}$)

The coefficient of volume expansion, β , and the coefficient of linear expansion, α , for the same material can be related through the following formula: $\beta = 3\alpha$



Common units of heat:

calorie (cal) – amount of heat needed to raise the temperature of 1 gram of water by 1°C .

kilocalorie (kcal) – amount of heat needed to raise the temperature of 1 kg of water by 1°C

British Thermal Unit (BTU) – amount of heat needed to raise the temperature of 1 lbm of water by 1°F

Heat is internal energy in transit from one body of matter to another by virtue of a temperature difference between them.

Conversion		
1 cal	4.186 J	0.00397 BTU
1 kcal	4186 J	3.966 BTU
1 BTU	252.16 cal	0.252 kcal
1 BTU	1055.06 J	1.055 kJ
1 J	0.239 cal	0.000239 kJ
1 J	0.000948 BTU	
Joule (J) is the SI unit of heat		

Heat Capacity

The amount of heat to be supplied to an object to produce a unit change in its temperature. Another term for heat capacity is Thermal Capacity. SI unit is J/K

$$Q = C \Delta T$$

$$C = \frac{Q}{\Delta T}$$

where:

C = heat capacity

Q = heat

 ΔT = change in temperature

$$Q = mc \Delta T$$

where:

c = specific heat

C = heat capacity

Q = heat

 ΔT = change in temperature

m = mass

$$c = \frac{Q}{m \Delta T} = \frac{C}{m}$$

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For monoatomic gases, $\gamma = 1.67$ For diatomic gases, $\gamma = 1.40$.

Molar specific heat capacity, c_m is the product of molecular weight and the specific heat of the substance.

$$C_m = (MW)(c)$$

Unit is BTU/lb-mol $^{\circ}\text{F}$ or J/kmol $^{\circ}\text{C}$

For gases:

$$Q = mc_v \Delta T$$

Constant volume process

$$Q = mc_p \Delta T$$

Constant pressure process

where:

 c_v = specific heat capacity at constant volume c_p = specific heat capacity at constant pressure

For gases c_p is always greater than c_v . The ratio is given through γ .

$$\gamma = \frac{c_p}{c_v}$$

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$$Q_L = \pm mL$$

where:

 Q_L = latent heat (J, cal, BTU)

m = mass of system (g or kg)

L = latent heat constant

For water at standard atm pressure

Latent heat	kg/kJ	cal/g	BTU/lbm
Fusion	333.5	79.7	143.4
Vaporization	2256.7	539.1	970.3
Sublimation	2838	667.8	1220

Sensible heat

It is the quantity of heat that changes the temperature of the substance without changing its phase.

The total heat, Q_T , entering a substance is the sum of the latent heat and the sensible heat.

$$Q_T = Q_L + Q_S$$

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Latent Heat

It is the quantity of heat absorbed or released by a substance undergoing a change of state, such as ice changing to water or water to steam, at constant temperature and pressure. Also known as heat of transformation. Latent heat of fusion – solid to liquid and vice versa
Latent heat of vaporization – liquid to gas and vice versa
Latent heat of sublimation – solid to gas without becoming liquid and vice versa